

MD. ROFAZ HASAN RAFIU

Generative AI • LLM Systems Engineer

Tangail, Dhaka, Bangladesh | +880-1794-678595 | mdrofazhasanrafiu@gmail.com | [linkedin.com/in/md-rofaz-hasan-rafiu](https://www.linkedin.com/in/md-rofaz-hasan-rafiu) | github.com/rofazhasan | <https://rofazhasan.github.io/rofaz-portfolio/>

PROFESSIONAL SUMMARY

Computer Science & Engineering student at RUET specializing in **Generative AI & LLM Systems**, **Local Small Language Model (SLM) Orchestration**, and structured inference pipelines. Experienced in designing low-temperature prompted evaluation engines (Ollama/Qwen/Llama), Zod schema validation, sub-10ms deterministic error fallbacks, and high-concurrency Next.js 15 / PostgreSQL architectures.

EDUCATION

- Rajshahi University of Engineering & Technology (RUET) — B.Sc. in Computer Science & Engineering**2023 – 2027
 - Admitted through national combined engineering entrance examination (CUET, KUET, RUET) ranking in top merit percentile.
 - Coursework: Algorithms & Data Structures, Object-Oriented Design, Database Systems, Computer Networks, Discrete Math.
- St. Joseph Higher Secondary School — Higher Secondary Certificate (HSC), Science**2022
 - GPA: 5.00 / 5.00

TECHNICAL SKILLS

- Generative AI & LLMs:** Local SLM Deployment (Ollama, Qwen 2.5, Llama 3.2), Prompt Engineering, Structured JSON Extraction, Zod / Schema Enforcement, Context Management, Low-Temperature Sampling, Hallucination Mitigation.
- AI Evaluation & Reliability:** Deterministic Fallback Pipelines, Groundedness Verification, Output Guardrails, Latency Optimization, Edge Inference Strategies, Error Logging & Diagnostic Telemetry.
- Backend & Inference APIs:** Python (FastAPI, Flask), TypeScript, Go (1.22+), Node.js (Next.js 15 App Router), PostgreSQL, Prisma ORM 6.10, Redis (Caching & Rate-Limiting), SQLite, RESTful APIs, WebSockets (Socket.io).
- DevOps & Platforms:** Docker, NGINX, GitHub Actions (CI/CD), Edge Caching (SWR, CDN Cache-Control), Linux / Bash Scripting, Git, Jest.

FEATURED GENAI & LLM PROJECTS

- Digital School AI Mistake Diagnostics Engine** [TypeScript, Next.js 15, Ollama, Qwen 2.5, PostgreSQL, Prisma]Live: rofazacademy.dev
 - Local SLM Inference Pipeline:** Designed and deployed an end-to-end Local SLM inference pipeline utilizing Ollama (qwen2.5:3b / llama3.2:3b) running low-temperature (0.3) system-prompted evaluations over student exam error vectors.
 - Structured Output Extraction:** Enforced strict markdown/JSON schema extraction parsing diagnostic reports into discrete data structures: performance overview, weak topic clusters, and 3-step action recovery plans.
 - Deterministic Fallback Algorithm:** Engineered a sub-10ms fallback routine computing topic error distributions and model explanations when local inference endpoints face latency spikes, ensuring 100% service uptime.
 - Bilingual Transliteration & Neural Conversion:** Integrated phonetic transliteration utilities in TypeScript to standardize English-Bangla academic tokens before model ingestion and dashboard rendering.
- FocusGuard — Policy Compilation & Enforcement Daemon** [Go 1.22, Docker, WebSockets, Chrome MV3, SQLite]github.com/rofazhasan/FocusGuard
 - Rule Compilation Engine:** Built a Go policy distributor compiling dynamic behavioral rules directly into native browser C++ filtering rules via Chrome Manifest V3 declarativeNetRequest for zero-overhead execution.
 - Distributed WebSocket Broadcast:** Maintained real-time synchronization of shared attention budgets across heterogeneous desktop and mobile clients.
 - Local-First Resilience:** Integrated SQLite offline policy caching with vector delta syncing; implemented Android VpnService UDP DNS sinkhole intercepting restricted domains with RFC 1035 NXDOMAIN.
- OMRView — Vision-Assisted Ground-Truth Parser** [React 19, TypeScript, Vite, OpenCV.js Wasm]github.com/rofazhasan/OMRView
 - In-Browser CV Ingestion:** Developed a client-side document rectification pipeline running compiled OpenCV WebAssembly to extract ground-truth answer vectors from camera captures, offloading 100% of compute from the server.
 - Geometric Normalization:** Applied 4-point homography perspective transformations to standardize physical test sheet inputs into structured JSON payloads for downstream automated grading.
- Study Companion — Multimodal LLM Learning Assistant** [Flutter, Dart, Firebase, Google Gemini AI, Kotlin]github.com/rofazhasan/study-companion
 - Gemini AI Integration:** Integrated Google Gemini API to generate structured academic summaries, quiz breakdowns, and contextual learning explanations.
 - Cross-Platform Telemetry:** Built with Flutter and Firebase real-time database to persist habit streaks and AI chat sessions across mobile devices.
- Atlas — macOS Native Operating System with Vision OCR** [Swift 6, SwiftUI, SwiftData, Apple Vision OCR, CryptoKit]github.com/rofazhasan/atlas
 - Engineered zero-knowledge AES-256 GCM encrypted vault with Apple Vision OCR receipt scanning and native SwiftData local persistence.
- Additional Open-Source Systems & Repositories**[GitHub: github.com/rofazhasan](https://github.com/rofazhasan)
 - BioEntry Pro** (github.com/rofazhasan/BioEntry): Biometric face verification attendance engine with TensorFlow.js WebGL/WASM acceleration in Electron.
 - Meal Manager V2** (github.com/rofazhasan/Meal-Management): Fullstack relational hostel ledger in React 18, TypeScript, and Prisma v5 PostgreSQL.
 - Krishi Bondhu AI** (github.com/rofazhasan/Krishi_Bondhu_AI): Offline-first agricultural diagnosis mobile application with SQLite queue.

- **sdms** (github.com/rofazhasan/sdms) & **FinTrack** (github.com/rofazhasan/FinTrack): Relational student records portal and mobile personal finance tracker.

ACHIEVEMENTS & LEADERSHIP

- **Creative Talent Hunt, Regional Champion:** Awarded by the Government of Bangladesh for demonstrated excellence in mathematics and computer science.
- **Technical Workshop Organizer:** Led technical peer workshops at RUET covering Data Structures, Algorithmic Complexity, and Object-Oriented Design.
- **Competitive Problem Solving:** Regular participant in university programming contests and regional hackathons.